

```

MariaDB [project]> CREATE TABLE Employee (
  ->     empNo VARCHAR(5) PRIMARY KEY,
  ->     fName VARCHAR(20),
  ->     lName VARCHAR(20),
  ->     address VARCHAR(50),
  ->     DOB DATE,
  ->     sex CHAR(1) CHECK (sex IN ('M','F')),
  ->     position VARCHAR(30) CHECK (
  ->         position IN ('Manager','Team Leader','Analyst','Software
Developer'))
  -> ),
  ->     deptNo VARCHAR(5)
  -> );
Query OK, 0 rows affected (0.015 sec)

```

```

MariaDB [project]> INSERT INTO Employee VALUES
  -> ('E1','Kofi','Mensah','Accra','1990-05-12','M','Manager','D1'),
  -> ('E2','Ama','Boateng','Kumasi','1992-07-22','F','Manager','D2'),
  -> ('E3','Yaw','Owusu','Cape Coast','1988-03-10','M','Analyst','D1'),
  -> ('E4','Akosua','Asante','Takoradi','1995-09-18','F','Team Leader','D3'),
  -> ('E5','Kwame','Appiah','Accra','1991-11-30','M','Software
Developer','D2'),
  -> ('E6','Efua','Antwi','Tamale','1993-01-25','F','Analyst','D3'),
  -> ('E7','Kojo','Bediako','Sunyani','1989-06-14','M','Software
Developer','D1'),
  -> ('E8','Abena','Osei','Ho','1994-12-05','F','Manager','D3');
Query OK, 8 rows affected (0.017 sec)
Records: 8  Duplicates: 0  Warnings: 0

```

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--
MariaDB [project]> CREATE VIEW FemaleManagerProjects AS
  -> SELECT p.projNo, p.projName
  -> FROM Project p
  -> JOIN Department d ON p.deptNo = d.deptNo
  -> JOIN Employee e ON d.mgrEmpNo = e.empNo
  -> WHERE e.sex = 'F';
Query OK, 0 rows affected (0.012 sec)

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MariaDB [project]> SELECT *
  -> FROM FemaleManagerProjects
  -> ORDER BY projNo;

```

projNo	projName
P2	Inventory System
P3	Banking App
P4	E-Commerce

```

CREATE VIEW EmpProjectDetails AS
  -> SELECT
  ->     e.empNo,
  ->     e.fName,

```

```

-> e.lName,
-> p.projName,
-> w.hoursWorked
-> FROM Employee e
-> JOIN WorksOn w ON e.empNo = w.empNo
-> JOIN Project p ON p.projNo = w.projNo;

```

Query OK, 0 rows affected (0.014 sec)

```

MariaDB [project]> SELECT *
-> FROM EmpProjectDetails;

```

empNo	fName	lName	projName	hoursWorked
E1	Kofi	Mensah	SCCS	20
E1	Kofi	Mensah	Inventory System	15
E2	Ama	Boateng	Inventory System	25
E2	Ama	Boateng	Banking App	30
E3	Yaw	Owusu	SCCS	10
E3	Yaw	Owusu	Banking App	20
E4	Akosua	Asante	SCCS	10
E4	Akosua	Asante	E-Commerce	35
E5	Kwame	Appiah	Inventory System	40
E5	Kwame	Appiah	Banking App	25
E6	Efua	Antwi	Banking App	30
E6	Efua	Antwi	E-Commerce	20
E7	Kojo	Bediako	SCCS	15
E8	Abena	Osei	Inventory System	10
E8	Abena	Osei	E-Commerce	40

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```

MariaDB [project]> CREATE VIEW EmpProject(empNo, projNo, totalHours)
-> AS
-> SELECT w.empNo, w.projNo, SUM(hoursWorked)
-> FROM Employee e, Project p, WorksOn w
-> WHERE e.empNo = w.empNo AND p.projNo = w.projNo
-> GROUP BY w.empNo, w.projNo;

```

Query OK, 0 rows affected (0.007 sec)

```

MariaDB [project]> SELECT *
-> FROM EmpProject;

```

empNo	projNo	totalHours
E1	P1	20
E1	P2	15
E2	P2	25
E2	P3	30
E3	P1	10
E3	P3	20
E4	P1	10
E4	P4	35
E5	P2	40
E5	P3	25

E6	P3	30
E6	P4	20
E7	P1	15
E8	P2	10
E8	P4	40

--

```
SELECT projNo
-> FROM EmpProject
-> WHERE projNo = 'SCCS';
Empty set (0.009 sec)
```

```
SELECT projNo
-> FROM EmpProject
-> WHERE projNo = 'P1';
```

projNo
P1
P1
P1
P1

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```
SELECT COUNT(projNo)
-> FROM EmpProject
-> WHERE empNo = 'E1';
```

COUNT(projNo)
2

--

```
MariaDB [project]> SELECT empNo, totalHours
-> FROM EmpProject
-> GROUP BY empNo;
```

empNo	totalHours
E1	20
E2	25
E3	10
E4	10
E5	40
E6	30
E7	15
E8	10

--

```

SELECT empNo, SUM(totalHours) AS totalHours
  -> FROM EmpProject
  -> GROUP BY empNo;

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empNo	totalHours
E1	35
E2	55
E3	30
E4	45
E5	65
E6	50
E7	15
E8	50

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7.28

to maintain this as a materialized view the materialized view must be maintained by handling insert, update operations to help manipulate and modify by adding and removing the partNo when and depending on if at least one row tuple satisfies the condition where partcost > 1000. also the base table must be checked to see if other qualifying rows exist. by maintaining the number of qualifying rows per partNo, the view can be kept and maintained incrementally without using and accessing the base table